Nishal Stanislaus Silva, *PhD*

Machine Learning Engineer | ML on Real Hardware, Real-Time Inference, Sensor & Time-Series Data | Applied ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

nishal.silva@hotmail.com | +1 613 200 2030 | Toronto, ON | nishal.xyz | linkedin.com/in/nishal-silva

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18th October 2026

Re: Senior / Staff Machine Learning Engineer, Vehicle Dynamics – Waabi

Dear Hiring Manager,

I am applying for the Senior / Staff Machine Learning Engineer, Vehicle Dynamics role on the Autonomy & Algorithms team at Waabi. I am a Machine Learning Engineer with a PhD (University of Trento, 2025) and 10+ years across industry and academia, and I build end-to-end ML systems that run on real hardware under hard latency limits. Waabi's approach of testing whether a learned method beats the classical baseline in closed-loop simulation and on the vehicle is the way I like to work: measure first, then decide.

Three parts of the posting match my work. For ML on a physical system, my published pattern detection system runs at 14 ms on a Raspberry Pi 4 at F1 0.76, 74x faster than the DTW baseline (Journal of the Audio Engineering Society, 2026), and I tested it live in front of audiences rather than only offline. For gesture sensing, I fused an LSTM on 1-second IMU windows with audio pattern detection through a hierarchical state machine (IEEE IS^2 2025). For closed-loop systems in industry, at MAS Holdings I integrated a camera-based care-label checker with a conveyor and PLC reject, and coordinated three closed-loop subsystems in custom C++ for a direct-to-garment printer. I also design the benchmark suites with baselines and ablations that show whether a new approach actually wins.

I should be direct about the gaps. My control experience is applied rather than theoretical: I have not built MPC, optimal control or vehicle dynamics models, and I have not done state estimation or system identification on a vehicle. My deep learning work is mostly TensorFlow and Keras, with PyTorch at a working level. What I bring is C++ and Python at production quality, linear algebra, signal processing and statistics, and a habit of owning a problem from offline experiments through to deployment on hardware. I expect to learn the vehicle control side quickly.

I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, and available immediately. I would welcome the opportunity to discuss how my experience can support the Autonomy & Algorithms team at Waabi. My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD